Application of Stormwater and Sediment Management Based on the ABC Waters Design Guidelines in the Upper Stream Suburban Section of the Mengkibol River, Kluang, Malaysia

Jiaxin Jiang

Abstract In the context of climate change, the increasing frequency of extreme and prolonged rainfall has significantly heightened the risk of urban flooding. Urban waterfront spaces are especially vulnerable, as floods often follow river channels, making adjacent zones primary targets of flood impact. Due to unreasonable zoning or limited urban land availability, these areas frequently lack sufficient space to function as floodable zones that regulate excess stormwater. Additionally, heavy rain-induced runoff can carry pollutants into rivers, threatening local ecosystems and public health. The scouring effect of floods on riverbanks also leads to increased sediment deposition, reducing channel capacity and exacerbating flood risks. Effective flood management must therefore incorporate sediment control strategies. However, current research often treats flood and sediment management as separate issues. This study adopts ABC Waters design guidelines, utilizing runoff routes and sediment transport index to develop an integrated flood retention and sediment management system for the upstream section of Kluang's Mengkibol River, under land constraints. The aim is to enhance the resilience and environmental quality of urban waterfront spaces against future climate uncertainties.

Keywords Resilient waterfront landscape design · Stormwater management · Sediment control · ABC waters · Kluang

1 Introduction

Stormwater management has become critical for most cities' journey toward achieving urban resilience. Under the influence of climate change, unpredictable rainfall patterns and extreme precipitation events have led to an increased frequency

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of urban flooding [1]. Concurrently, urbanization, through ongoing land development and infrastructure expansion, has significantly increased the area of impervious surfaces within cities [2]. This process has substantially altered the response of urban hydrological and river systems to the built environment [3]. Driven by climate change and urbanization, the stormwater load on urban drainage systems has markedly intensified, presenting significant challenges to city stormwater management [4, 5].

In urban riverfront areas, flooding issues tend to be more frequent and severe. Rivers serve as the natural drainage network within cities. However, the limited capacity of these waterways often makes it challenging to accommodate high-intensity, prolonged rainfall, leading to flooding in riverfront regions. Inadequate planning during the early stages of urban development or the scarcity of urban land has often resulted in insufficient flood buffers along riverbanks. Consequently, built-up areas adjacent to the river become direct targets of flooding, causing significant economic losses, casualties, and burdens of environmental governance. Therefore, implementing targeted stormwater management measures in urban riverfront areas is critical.

Simultaneously, the issue of pollutant diffusion caused by river floods carrying sediments, which threaten public health, should be considered. Sediments are solid particles transported by water flow that settle on riverbeds, banks, or near river mouths. They typically consist of silt, gravel, stones, and organic matter. Sediments are a crucial component of river ecosystems, serving as a nutrient-rich reservoir that influences the exchange of materials within the river [6]. However, riverbed sediments can also act as a repository for accumulated pollutants [7]. Under the influence of water flow, sediments can be transported, leading to changes in riverbed morphology and the movement of solutes within the riverbed. Research indicates that during peak flow events, a significant portion of the biological load may originate from riverbed sediments [8, 9]. Floodwater flow instability significantly affects the movement of sediment particles in the riverbed, thereby influencing the diffusion of pollutants [7]. Then, it can negatively impact downstream river ecosystems and contribute to the spread of diseases.

Moreover, the increased rainfall associated with climate change can lead to a higher discharge of sediments [10]. The accumulation of large amounts of sediment can alter the elevation and cross-sectional geometry of riverbeds [11–15], reducing the water-holding capacity of river channels, weakening their flood storage ability, and ultimately exacerbating flooding [16, 17]. Therefore, sediment management is a critical component of stormwater management. During the early stages of planning and design, practitioners should comprehensively consider both aspects to ensure urban safety, public health, and river ecosystem protection during heavy rainfall events. However, sediment transport and management are often overlooked in flood risk studies and stormwater management practices [18]. Stormwater management and sediment management are frequently treated as two separate issues.

Gray infrastructure represents the traditional approach to stormwater management, primarily focusing on collecting, storing, and conveying stormwater to receiving water bodies [19, 20]. During the last century, urbanization led to significant transformations in river environments. Rigid, artificial channels replaced natural

drainage channels to facilitate the development of impervious surface areas [21]. With the continuous improvement and expansion of urban underground drainage networks, structural measures gradually became the dominant means of urban drainage. However, traditional drainage systems have a fixed capacity and tend to deteriorate over time, leading to a decline in performance.

Moreover, the construction and upgrading of drainage facilities require substantial financial investment. Consequently, conventional gray infrastructure struggles to cope with the increasingly severe and unpredictable risks posed by extreme and prolonged rainfall events. As a result, there has been a gradual shift in approaches and strategies for urban stormwater management. Over the past two decades, more sustainable and resilient stormwater management methods have been proposed worldwide, such as implementing blue-green infrastructure. Examples include Stormwater Control Measures (SCM) in the United States, Water Sensitive Urban Design (WSUD) in Australia and New Zealand, Low Impact Development (LID) in China, Sustainable Urban Drainage Systems (SUDS), and Nature-Based Solutions (NBS) in Europe. These blue-green infrastructure approaches often complement existing urban gray infrastructure to achieve more effective, sustainable, and integrated stormwater management [22]. However, these methods focus more on enhancing stormwater retention capacity, with relatively less emphasis on water quality improvement.

Historically, sediment management in river basins and waterways has been a critical issue for water resource managers [23]. Current research on river sediment primarily focuses on sediment transport in large basins, such as the Nile and Yellow River basins. Sediment management efforts are primarily concentrated on reservoirs or open basins, with specific measures typically involving using rigid structures to intercept sediments. In contrast, relatively little research is dedicated to sediment management in urban river channels during flooding.

This study identifies the limitations of existing approaches to stormwater management and sediment control, and explores the application of an integrated methodology within the specific site. Traditionally, stormwater management and river sediment control have been treated as two separate domains in waterfront planning and management, lacking systemic and coordinated solutions. In response, this research proposes an integrated design strategy grounded in Singapore's ABC Waters Design Guidelines, which incorporates both stormwater and sediment management within a unified spatial framework. By emphasizing the interrelationship and synergetic regulation between these two aspects, the study offers a replicable case of practice that contributes to the ecological resilience of urban riverfronts and the advancement of holistic water environment governance.

2 ABC Waters Design Guidelines

Singapore is a typical tropical city with high annual rainfall and limited land resources, posing significant challenges for urban stormwater management systems. Urbanization has dramatically altered the island's water balance [24]. However, given the constraints on land availability, expanding drainage channels to increase the capacity of the stormwater network is not a feasible option. In response, Singapore's national water agency, PUB, launched the Active, Beautiful, Clean (ABC) Waters program in 2006 [25]. The program aims to achieve sustainable stormwater management, quality improvement, and enhanced aesthetic and recreational value of water environments by 2030 [24].

To promote the implementation of this initiative, the government has partnered with developers to establish an integrated stormwater management strategy as outlined in ABC Waters, which comprises source, pathway, and receptor components. The 'source' refers to the origin of stormwater runoff, emphasizing on-site collection and attenuation of runoff through ABC design features. The 'pathway' represents the route through which stormwater is conveyed, focusing on enhancing the capacity of the stormwater transport system, including measures like widening and deepening channels and detention of catchment areas. The 'receptor' denotes locations where stormwater accumulates and may impact infrastructure, emphasizing the arrangement of protective measures. On a micro level, ABC Waters proposes six treatment elements to achieve purification, detention, retention, conveyance, and infiltration. The design features corresponding to these five primary functions are listed in the Table 1.

Although the treatment elements in ABC Waters share similarities with stormwater management methods popular in other countries and regions [26], the design features of ABC Waters place a greater emphasis on water quality improvement [27]. Specifically, it focuses on purifying stormwater runoff and river pollutants, thereby contributing to sediment management by reducing the pollutant content within sediments. While numerous ABC Waters landscape projects have already been implemented within Singapore, this approach has yet to be widely adopted in other tropical cities with similar environmental conditions.

Singapore's northern neighbor, Malaysia, shares similar environmental conditions in certain areas. Flooding is one of the significant environmental challenges facing Malaysia. From November to March each year, the northeast monsoon brings heavy rainfall, which frequently causes widespread flooding. Despite the local government's efforts to implement several initiatives to mitigate flooding, traditional measures have proven insufficient to meet urban and rural areas' long-term flood control needs [28]. Therefore, this study proposes to select Kluang, a city in Johor, Malaysia, that suffers from flood and river pollution, as a pilot city for promoting the ABC Waters design guidelines. Utilizing the ABC Waters design principles as the methodology and a systematic landscape design approach as the outcome, this project aims to provide a reference case for stormwater management and sediment

Table 1 Functions of ABC waters design features (self-drawn)

	Vegetated swale	Bioretention swale	Bioretention basin (rain garden)	Sedimentation basin	Cleansing biotope	Constructed wetland
Design feature	They are natural drainage channels with gentle slopes that slow stormwater flow, promoting sediment settlement and reducing erosion. Often used alongside bioretention systems	They are vegetated channels with integrated bioretention systems. They temporary ponding to enhance infiltration. Runoff is filtered as it seeps through, with cleansed water collected by subsoil pipes	They are vegetated channels with integrated bioretention systems. They temporary ponding to enhance infiltration. Runoff is filtered as it seeps through, with cleansed water collected by subsoil pipes	These ponds temporarily detain stormwater, slowing flow for particle settling, and often serve as pre-treatment for wetlands or as water storage and irrigation ponds	Cleansing biotopes are vertical flow wetlands with recirculation, using nutrient-poor substrates and wetland plants for their water-purifying abilities	They are shallow, vegetated waterbodies used for stormwater management. They consist of an inlet zone for sediment removal, a macrophyte zone for pollutant filtration, and a high-flow bypass channel
Function	🔵 🟡 🟢	🔵 🟡 🟢 🔴	🔵 🟡 🔴	🔵 🟡 🟣	🔵 🟡 🔴	🔵 🟡 🟣

🔵 Purification, 🟡 Detention, 🟣 Retention, 🟢 Conveyance, 🔴 Infiltration

control in the riverfront areas of Kluang, transforming it into a more resilient and livable city.

3 Study Method and Study Area

3.1 Study Method

In this study, the author employed a series of software tools to visualize the preliminary analysis and design practices. The software used includes Adobe Photoshop 2021, Adobe Illustrator 2021, Rhino 7.0 with Grasshopper, and ArcGIS Pro.

Specifically, Rhino 7.0 with Grasshopper and ArcGIS Pro were integrated with DEM data to generate terrain models and conduct analyses of surface runoff direction and flood extent. Based on the 3D terrain model and satellite imagery, Rhino 7.0 was further utilized for environmental modeling, the drawing of existing and proposed river sections, as well as the development of axonometric diagrams to illustrate design strategies. Finally, the Adobe software suite was used to enhance vector drawings, incorporate raster imagery, and arrange the overall layout, resulting in a clearer and more visually appealing presentation.

3.2 Kluang and the Mengkibol River

Kluang is a city located in the central region of Johor, Malaysia, covering a total area of approximately 28,518.8 km^2, with a council administrative area of about 126.57 km^2 [29]. In 2023, the city had an estimated population of around 328,000 [30]. The primary industries in Kluang include agriculture, manufacturing, and tourism. Kluang experiences a tropical rainforest climate, with an average annual temperature of 25.6 °C and an average annual rainfall of 2,473 mm [31]. Influenced by the northeast and southwest monsoons, Kluang's climate can be categorized into a wet season (from November to March) and a less wet season (from April to October). The city is in a hilly region, with an elevation range between −1 and 941 m above sea level, the highest point being Gunung Lambak in the east. Despite numerous small hills and undulating terrain within Kluang, there are also substantial areas of flat lowlands. While these geographical features provide favorable conditions for agricultural activities, they also pose flooding risks (Fig. 1).

The river system in the central area of Kluang is primarily composed of the Sungai Melantai in the west and the Mengkibol River in the east. Both rivers flow from south to north through the center of Kluang and converge in the northern area to form the Sungai Melantai, which eventually drains into the Sungai Sembrong. The Mengkibol River is the mother river of Kluang, extending approximately 8.5 km from south to north. Based on the functional attributes of the buildings along its banks, the

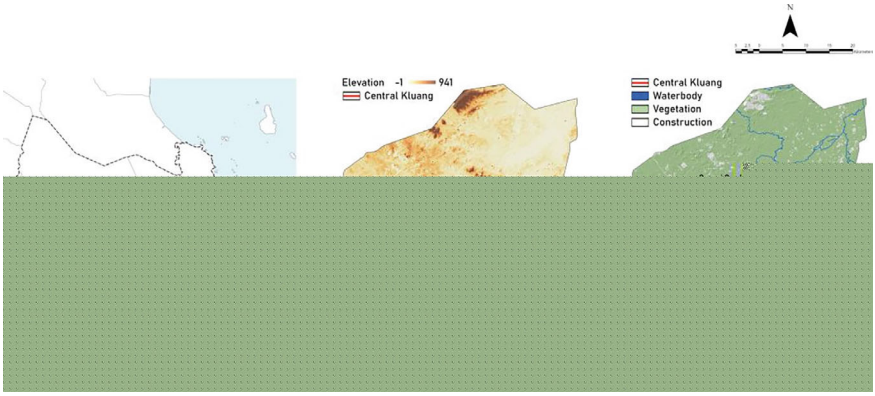

Fig. 1 Location, topography, and river system of Kluang (self-drawn)

Mengkibol River can be divided into three segments: (1) the upper suburban section, characterized by the presence of villages, industrial parks, and oil palm plantations; (2) the urban core, dominated by commercial and residential buildings; and (3) the lower suburban section, where oil palm plantations are predominant, with a few scattered villages (see Fig. 2). The morphological features of the Mengkibol River include a relatively narrow channel width, ranging from approximately 10 to 25 m, and steep banks with a slope of about 45°. Under climate change's impact, flooding along the banks of the Mengkibol River has become increasingly severe.

Fig. 2 Flooding extent of Kluang and flooding sections of the Mengkibol River in the urban context (upstream and midstream) (self-drawn)

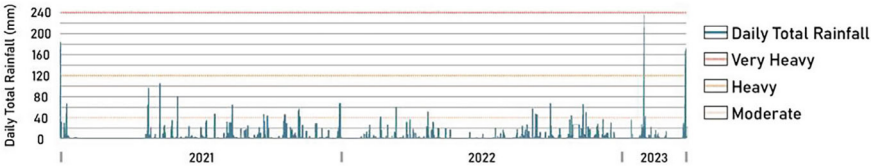

Fig. 3 Rainfall of Kluang (2021–2023) (self-drawn)

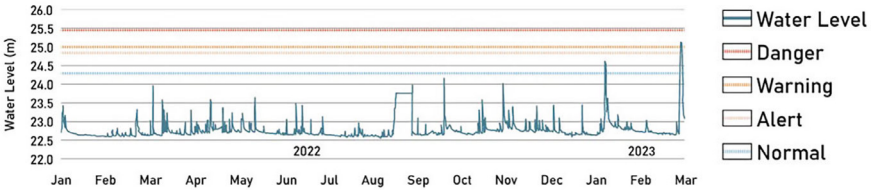

Fig. 4 Water level of the Mengkibol River (2022–2023) (self-drawn)

3.3 Flooding in Kluang

In recent years, the annual rainfall in Kluang has become increasingly unpredictable, with a higher probability of extreme peaks (see Fig. 3). As a result, the maximum water levels of the Mengkibol River during the rainy season have been steadily rising [32], exacerbating the risk of flooding in the surrounding areas (see Fig. 4). In March 2023, Kluang experienced another continuous heavy rainfall, leading to widespread flooding across the city and affecting 1,952 people [33]. The average monthly rainfall reached 209 mm, and the river's water level exceeded the warning level by 0.3 m. The extent of flooding in Kluang increased progressively from the upper to the lower reaches of the river. According to GIS calculations, the total flooded area was 1,847 ha. Of the affected buildings, 77% were residential, 12% were commercial, and 11% served other functions (see Fig. 3).

In recent years, with the development and construction along the banks of the Mengkibol River, the buffer zone area that can accommodate flooding has gradually decreased. Additionally, due to the river's narrow and elongated shape, its capacity for flood retention is significantly limited. These two factors have led to a lack of sufficient blue-green infrastructure in the city to regulate such large volumes of rainfall, causing severe impacts on the local community and economy.

3.4 Sediment Issue Along the Mengkibol River

In addition to flooding, sediment accumulation is another issue for the Mengkibol River. Observations from site investigations indicate that the amount of sediment and

Fig. 5 Sediment and insoluble solid waste along the Mengkibol River (self-drawn)

insoluble solid waste increases progressively from the upstream to the downstream sections of the river (see Fig. 5). Interviews with residents revealed that riverbank development has led to surface runoff carrying various types of pollutants into the river, contributing to increased sediment levels. This process not only results in river pollution but also causes the riverbed to rise, ultimately reducing the river's flood retention capacity and exacerbating flooding along its banks.

This project addresses the issue of sedimentation by utilizing Digital Elevation Model (DEM) data (https://earthexplorer.usgs.gov/). The author employed Grasshopper and GIS to visually represent the surface runoff directions, the Sediment Transport Index (STI), and sediment sources along both sides of the river. Surface runoff illustrates the natural flow paths of rainwater as it carries surface materials across the land. The STI indicates the potential for soil erosion within a specific area; higher STI values suggest a greater ability for sediment transport and a higher likelihood of soil erosion (see Fig. 6). The sources of river sediment can be categorized into four types: construction, agriculture, commercial activities, and erosion (natural or due to flooding). These sources lead to six river pollutants, resulting in a range of negative impacts (see Fig. 7).

4 Application of the Cleansing Sponge Design Framework Based on the ABC Waters Design Guidelines

4.1 Design Area

The site is on the upper west side of the Mengkibol River. On the western side of the river, there are orchards, the rail corridor, and the golf course, while a village occupies the eastern side. The reasons for selecting this site include: (1) It is one of the few open spaces along the riverbank, making it a promising candidate for conversion into blue-green infrastructure with stormwater and flood regulation functions. (2) Being situated upstream, implementing sediment control measures at this site could help reduce sediment accumulation downstream. (3) The land is proposed as residential in the Kluang zoning plan. However, the village across from the site has been severely affected by flooding. If this area is developed into a residential zone in the future, it may also face flood risks, potentially compromising the residents'

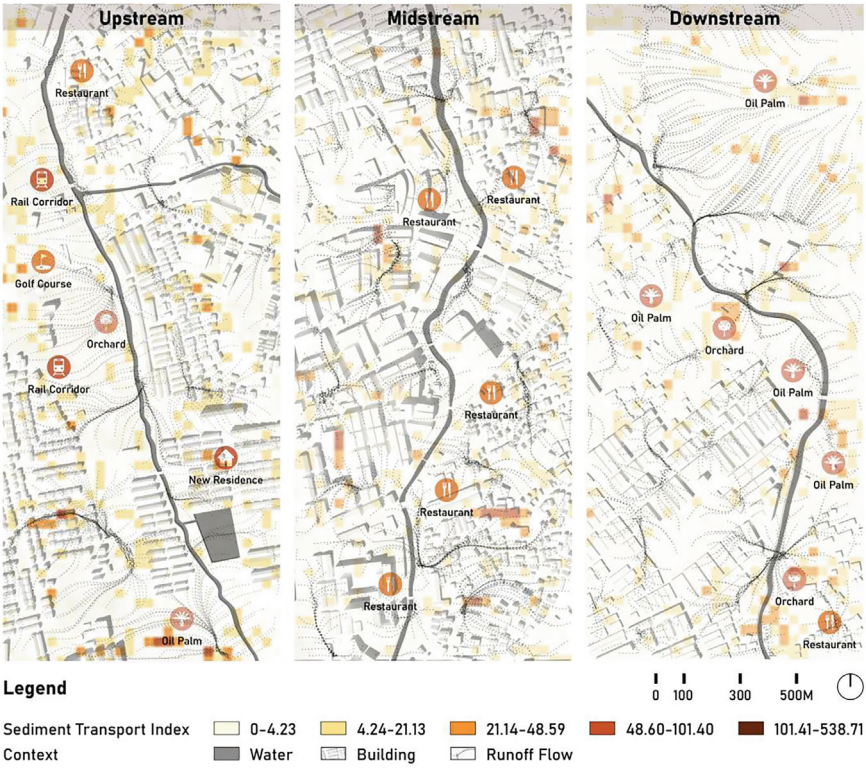

Fig. 6 Sediment transport index along the Mengkibol River, the distribution of possible pollution sources and the surface runoff flow (self-drawn)

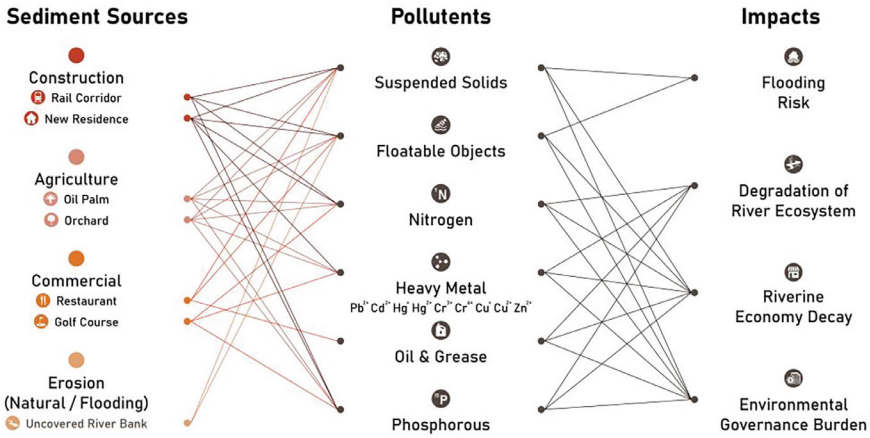

Fig. 7 Sediment sources, pollutants, and their negative impacts on the river (self-drawn)

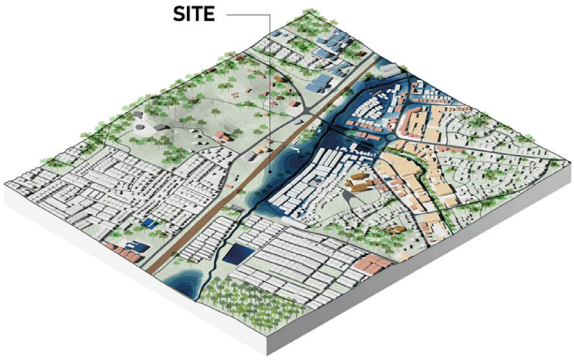

Fig. 8 Flooding scenario simulation of the site (self-drawn)

rights and negatively impacting real estate investment returns (see Fig. 8). Therefore, the author selected this site for intervention, guided by the ABC Waters Design Guidelines principles, to alleviate flooding and sediment issues within the area.

4.2 Three Strategies

Sponge Strategy. This strategy has two primary objectives: first, to increase the water storage capacity of the river channel, and second, to create swales to increase stormwater runoff routes (see Fig. 9). The ultimate goal is to reduce the likelihood of flooding on both sides of the riverbanks. To increase the river channel's water storage capacity, the author proposes three strategies: creating meandering streams along the river channel (S-1-1); widening the river channel (S-1-2); and installing underground water tanks in low-lying areas adjacent to the river while maintaining the above-ground space as a public activity area (S-1-3). When the river exceeds the warning level, it can flow into the underground water tanks. Meanwhile, to increase stormwater runoff routes by creating vegetated swales (S-2-1), rain gardens (S-2-2), and bioretention swales (S-2-3), thereby extending the time it takes for runoff to reach the river channel and delaying the river's peak volume.

Purification Strategy. This strategy aims to reduce sedimentation due to erosion during flooding and purify pollutants in stormwater runoff (see Fig. 10). Riverbanks are the areas most prone to soil erosion, necessitating bank stabilization measures to prevent erosion during flood periods. The erosion can lead to excessive soil entering the river channel, forming sediment, and raising the riverbed. Initially, based on the bank's slope, three stabilization measures will be implemented: vegetation, terraces, and gabions (C-1-1, C-1-2, C-1-3). As for the purification, in the proposed swales along the riverbanks, wetland plants will play a role in adsorbing and purifying pollutants from the runoff, thereby reducing the entry of pollutants into the river and their attachment to sediment. The three main approaches to existing sediment are to capture and remove it. Sediment will be collected using sediment basins (C-2-1),

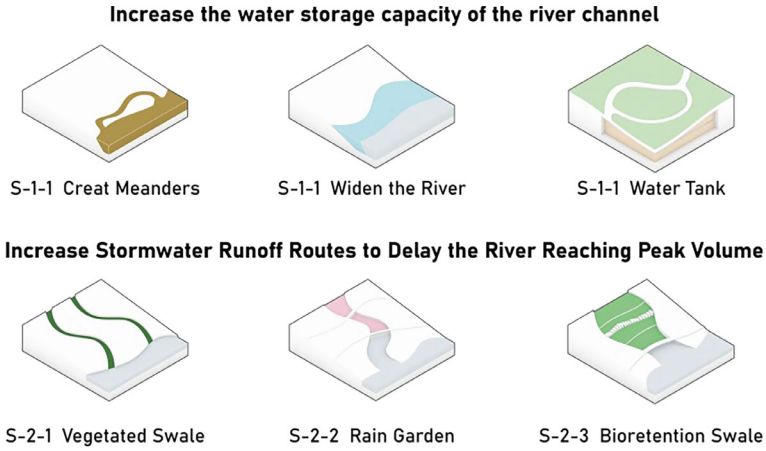

Fig. 9 Sponge strategy (self-drawn)

topo capturers (C-2-2), and wetland plants (C-2-3). Once the sediment is captured, it will be removed periodically using mechanical means (C-3-1), artificial means (C-3-2), or a combination of both (C-3-3).

Recreation Strategy. This strategy aims to establish a connection between the site and the surrounding residents, meeting their needs for access and recreational activities during both the wet and normal seasons (see Fig. 11). The recreational space can be categorized into point, linear, and surface-based. Point-based spaces include the bridge (R-1-1), water steps (R-1-2), and viewing steps (R-1-3). Linear spaces consist of riverbank trails (R-2-1), slope trails (R-2-2), and water trails (R-2-3). Area-based spaces comprise the amphitheater (R-3-1), event lawn (R-3-2), and sunken playground (R-3-3). The latter two are designed as sunken spaces that serve as open green areas for recreational activities during normal conditions and water storage areas during the wet season.

4.3 Master Plan Deduction

Firstly, the lowest points within the site are selected to serve as the underground water tank or sunken activity spaces, with three low points identified. Since the central and southern points are closer to the village and offer more significant potential for interaction with residents, these two points are designated sunken activity spaces, specifically the event lawn in the central area and the sunken playground in the southern area. The northernmost point is utilized as an underground water tank (Step 1). Additionally, the existing river channel is widened to the west, and a meander is created in the central section to enhance the overall water retention capacity of the river channel (Step 2).

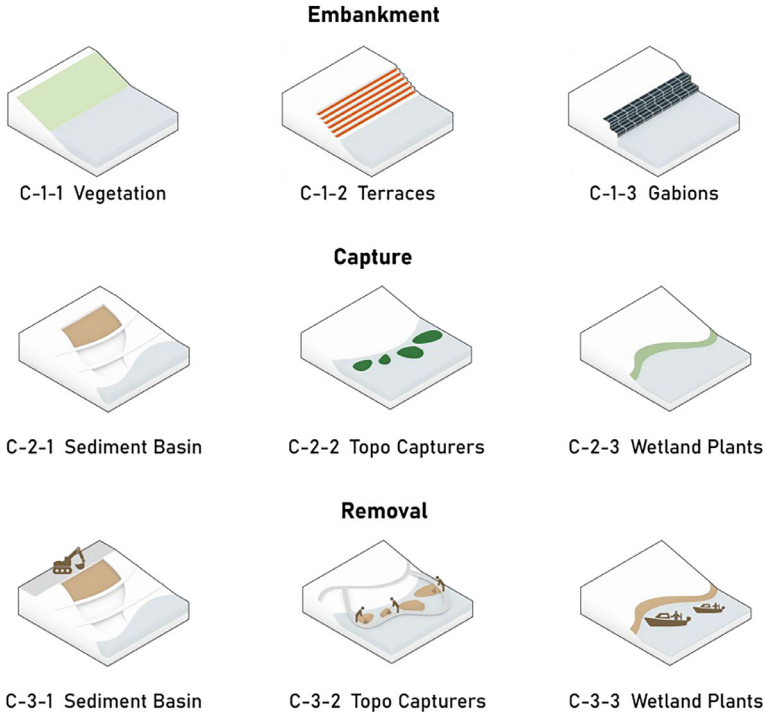

Fig. 10 Purification strategy (self-drawn)

The second strategy involves applying different bank stabilization measures based on the slope gradient to reduce sediment caused by riverbank erosion (Step 3). Specifically, vegetation ($10° < \text{slope} \leq 20°$), terracing ($20° < \text{slope} \leq 30°$), and gabion ($\text{slope} > 30°$) stabilize the riverbank. Vegetative cover is primarily arranged on the western side of the river channel, where the slope is relatively gentle and can form a floodplain area during the wet season. Terracing is positioned along the eastern riverbank, bordering the village, where it stabilizes the bank and serves as a recreational area for residents, enhancing the riverside landscape quality. In the northern and southern sections of the eastern bank, gabion are used for stabilization, providing lower interactivity but higher stability on steeper slopes.

The third strategy focuses on sediment capture (Step 4). Areas on the site's western boundary with high STI values, which indicate a higher risk of erosion, are transformed into gently sloped sedimentation ponds to capture external sediment sources, thereby reducing soil erosion in this region. Following this, swales are designed to allow wetland plants along the banks to perform a secondary capture of residual sediment, further minimizing external sediment input into the river. Additionally, based on fluvial dynamics, the flow velocity is slower along the concave bank of the river, making it more conducive to sediment deposition. Therefore, topo capturers are positioned along the concave bank to trap sediment already present in the river,

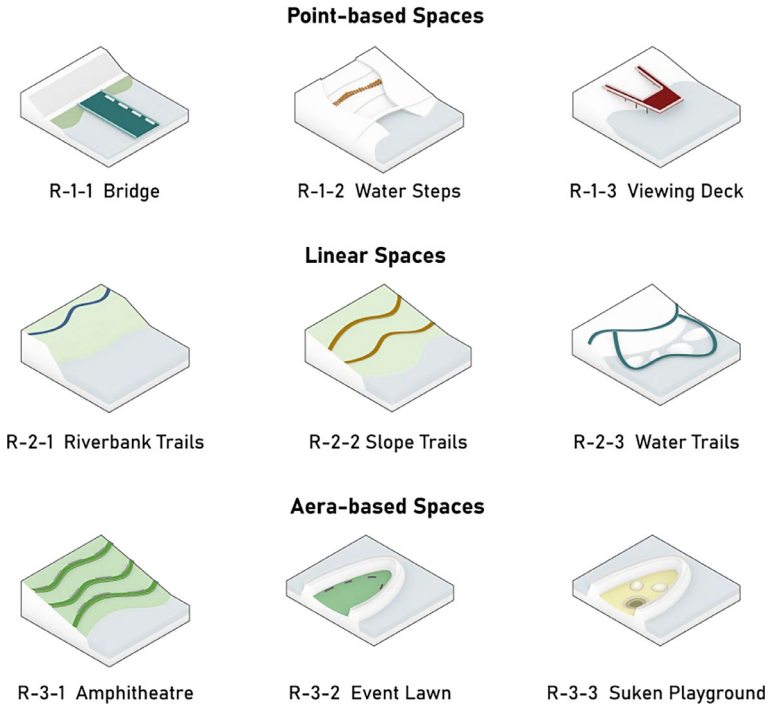

Fig. 11 Recreation strategy (self-drawn)

with wetland plants on the western side also contributing to sediment capture within the river channel (Fig. 12).

The fourth strategy involves arranging a swale system based on surface runoff flow lines to purify the runoff entering the river (Step 5). Since different types of swales have varying capacities for stormwater management, specific swales are positioned according to the concentration of surface runoff. Vegetated swales are placed in

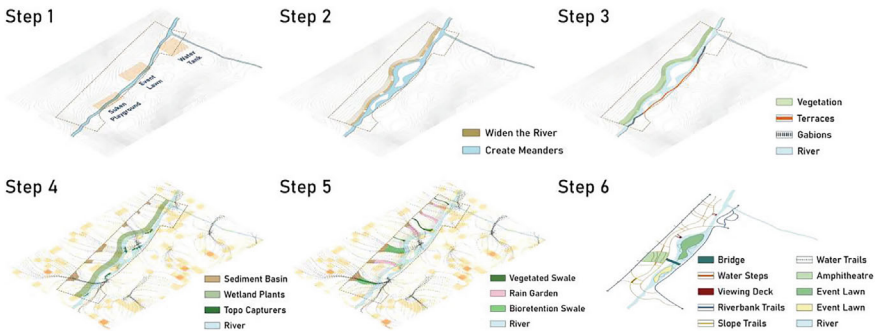

Fig. 12 Master plan deduction (self-drawn)

areas with low runoff concentration, rain gardens in areas with moderate runoff concentration, and bioretention swales in areas with high runoff concentration.

The fifth strategy addresses the placement of various recreational spaces (Step 6). Riverbank, slope, and water trails are arranged sequentially along the contour lines to provide different levels of riverside walking and water interaction experiences. Rain gardens, bioretention swales, and water steps enhance the site's interactivity and spatial interest. A bridge in the central area connects the eastern village with the western amphitheater, strengthening connectivity between the two banks. Water steps link the riverbank to the sunken spaces. The larger sunken space in the central area is designated as an event lawn, providing a venue for residents to host everyday outdoor activities and weekend markets. The smaller sunken space in the southern area is mainly designed as a children's play area (Fig. 13).

4.4 Three Systems

Sponge System. During the wet season or extreme rainfall events, the system efficiently guides runoff and stores water, thereby reducing the risk of flooding along the riverbanks. The functional modules of the sponge system consist mainly of topographical features and wetland vegetation (see Fig. 14), where the former directs water flow, and the latter assists by enhancing soil water infiltration through its root system. When the river reaches a high water level, the vegetated gentle slope area and sunken spaces on the left bank will serve as floodable zones. The vegetation in these floodable areas is designed to be flood-tolerant and has stable root systems to withstand the impacts of inundation during flood events (see Fig. 15).

Purification System. The primary components of this system are wetland plants with water purification capabilities, which function throughout the year to reduce the accumulation of pollutant-laden sediments in the river channel. Plant selection is based on the analysis of pollutant sources mentioned earlier, and their arrangement is determined by their flood tolerance in a vertical layout. During the dry season, runoff from the site's western side flows through vegetated swales, rain gardens, or bioretention swales, allowing the plants to absorb or capture pollutants. In the wet season, the swale system within the floodable areas helps to retain the runoff, with wetland plants purifying the water. The introduction of wetland plants not only improves water quality, enhances landscape aesthetics and increases biodiversity within the site (see Fig. 16).

Recreation System. This system comprises a series of landscape and recreational facilities designed to promote interaction between the surrounding residents and the site. During the dry season, these recreational facilities not only meet the needs of the residents for various daily activities but also serve as venues for festive events. For example, the event lawn can host weekend markets and traditional local performances, while the amphitheater can serve as a space for spectators and everyday relaxation. The landscape features in this area primarily consist of pathways and

Fig. 13 Master plan (self-drawn)

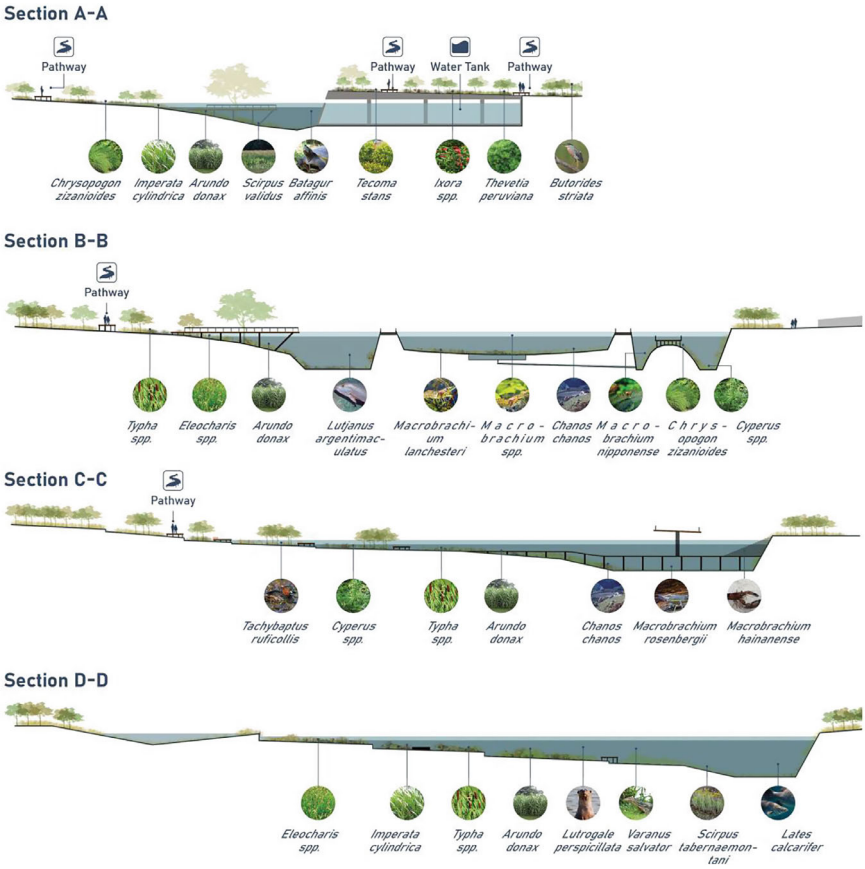

Fig. 14 Rainy season section (self-drawn)

viewing decks to minimize damage from flooding. Most of the pathways within the floodable areas will be submerged during heavy rainfall. However, the pathways near the site's western boundary are situated above the warning water level, ensuring pedestrian accessibility around the site during such events (Fig. 17).

5 Discussion

ABC Waters is an exemplary systemic stormwater management program replicating the natural water cycle within urban environments by integrating engineering techniques and ecological knowledge. As discussed earlier, the significant challenges in urban stormwater management are primarily driven by the intensification of climate change, urbanization, and the constraints posed by traditional grey infrastructure.

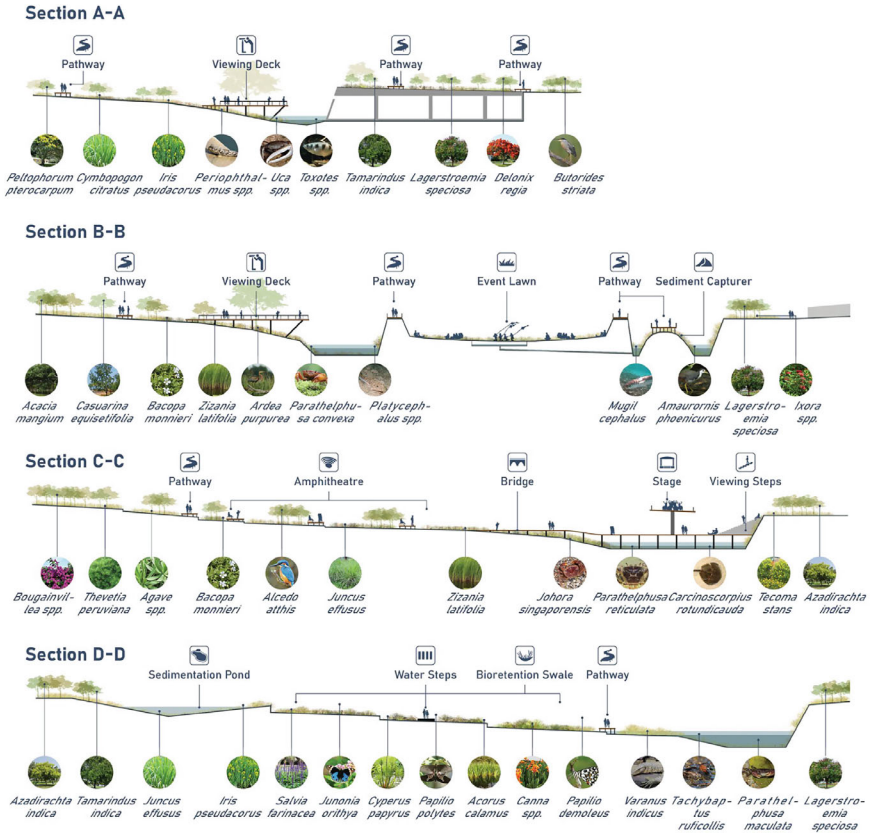

Fig. 15 Less wet season section (self-drawn)

Existing solutions are increasingly inadequate to address these escalating issues at their source. In response, ABC Waters proposes a series of more resilient and efficient blue-green infrastructure solutions, such as bioretention swales, cleansing biotopes, and constructed wetlands, to tackle the unpredictable challenges of urban stormwater. This design approach also emphasizes water quality improvement through wetland vegetation and topographical modifications, reflecting a comprehensive consideration of stormwater and sediment management during heavy rainfall events. Furthermore, ABC Waters underscores the interaction between artificial water systems and surrounding communities and the promotion of environmental education and awareness. These elements highlight its distinction from water management strategies proposed by other countries and regions.

However, ABC Waters is not without controversy. Regarding sediment management, some experts have argued that compared to traditional complex engineering channels, the softscape elements of ABC Waters may lead to increased sediment accumulation in waterways. Indeed, the soft banks, primarily composed of wetland

Vegetation Species	Flood Tolerance	Planting Location	Pollutent Purification Performance									
			Suspended Solids	Nitrogen	Heavy Metal						Oil & Grease	Phosphorous
					Pb^{2+}	Cd^{2+}	Hg^{2+}	Cr^{6+}	Cr^{4+}	Cu^{2+}	Zn^{2+}		
Cyperus spp.	Strong	Below Low Water Level	√	√									√
Scirpus validus	Strong	Low to Middle Water Level	√	√	√					√	√		√
Scirpus tabernaemontani	Strong		√	√						√	√		√
Juncus effusus	Strong	Middle to High Water Level	√	√	√					√	√		√
Cyperus papyrus	Strong		√	√		√	√			√			√
Acorus calamus	Strong		√	√	√	√				√	√	√	√
Canna spp.	General		√	√					√	√		√	√
Zizania latifolia	General		√	√	√	√	√	√	√	√	√		√
Arundo donax	Strong		√	√	√		√						√
Iris pseudacorus	Strong	High to Warning Water Level	√	√						√	√	√	√
Bacopa monnieri	Strong		√	√	√		√			√	√		√
Chrysopogon zizanioides	General		√	√	√					√			√
Imperata cylindrica	General		√	√						√	√		√
Typha spp.	Strong		√	√	√	√	√		√	√	√	√	√
Eleocharis spp.	Strong		√	√	√					√			√
Peltophorum pterocarpum	Weak	Above Warning Water Level	/										
Tamarindus indica	Weak												
Lagerstroemia speciosa	General												
Delonix regia	Weak												
Acacia mangium	Weak												
Casuarina equisetifolia	Weak												
Ixora spp.	Weak												
Bougainvillea spp.	Weak												
Azadirachta indica	General												
Cymbopogon citratus	General												
Thevetia peruviana	Weak												
Agave spp.	Weak												
Tecoma stans	General												
Salvia farinacea	Weak												

Flood Tolerance ● Strong ● General ● Weak

Fig. 16 Planting list and features of the selected plants (self-drawn)

Provision of Habitats for					
Fish	**Crustaceans**	**Amphibians**	**Birds**	**Reptiles**	**Insects**
Chanos chanos	*Carcinoscorpius rotundicauda*	*Lutrogale perspicillata*	*Alcedo atthis*	*Batagur affinis*	*Junonia orithya*
Lates calcarifer	*Johora singaporensis*		*Amaurornis phoenicurus*	*Varanus indicus*	*Papilio demoleus*
Lutjanus argentimaculatus	*Macrobrachium hainanense*		*Ardea purpurea*	*Varanus salvator*	*Papilio polytes*
Mugil cephalus	*Macrobrachium lanchesteri*		*Butorides striata*		
Periophthalmus spp.	*Macrobrachium nipponense*		*Tachybaptus ruficollis*		
Platycephalus spp.	*Macrobrachium rosenbergii*				
Toxotes spp.	*Macrobrachium spp.*				
	Parathelphusa convexa				
	Parathelphusa maculata				
	Parathelphusa reticulata				
	Uca spp.				

Fig. 17 Increased biodiversity in the proposed site (self-drawn)

vegetation, soil, and gravel layers, may contribute to more erosion-related sediment than hardened riverbanks. Nevertheless, these soft banks have the potential to filter runoff from adjacent areas, reducing the likelihood of pollutant-laden sediments entering the waterways. Additionally, appropriate sediment capture measures can be implemented within the water channels. For instance, in the design practices of this study, topo capturers are installed on convex banks to collect sediments, with periodic removal of excess sediment. Furthermore, GIS-based calculations of the site's Sediment Transport Index (STI) can guide the implementation of complex engineering measures, such as gabions and terraces, in areas prone to erosion. By integrating complex engineering solutions with the soft structures of ABC Waters, a more comprehensive strategy for addressing riverbank erosion and flood control can be developed.

Although ABC Waters provides a refined approach to managing urban stormwater runoff, managing riverine flood events and sediment control remains an issue that requires consideration from a watershed-level perspective. Consequently, there are certain limitations within the design guidelines. In other words, if individual sites are not defined at a macro level in terms of their specific roles in stormwater management and sediment control, and if the guidelines are applied indiscriminately, the design features may not achieve their maximum performance. Therefore, the ABC Waters design guidelines should be further developed to include broader watershed-level flood and sediment management considerations to reflect their systematic and intelligent approach better. The primary shortcoming of this study's design is the need for a more macro-level perspective on flood and sediment issues. As a result, the proposed design can only address localized problems to a limited extent and cannot achieve comprehensive watershed management. Flood risks and river pollution will likely persist. Nonetheless, this design offers a practical exploration of applying the ABC Waters guidelines outside Singapore, providing a reference case for stormwater and sediment management in riverfront areas.

Moreover, the limitations of ABC Waters are also evident in that its strategies primarily address design and maintenance without sufficient early intervention and guidance at the urban planning policy level. Proactive measures in policy can better equip cities to face the challenges of climate change. The site in this study exemplifies a case where insufficient consideration was given to flood-prone riverbank areas during the initial planning stages. Consequently, as urban development continues along the riverbanks, the latent risks embedded in the early planning phase have become more apparent, ultimately resulting in flooding. The generation of significant levels of pollutants in the river is also closely linked to management policies. Different types of establishments should be required to treat and monitor the quality of their site-generated runoff before discharging it into the river, ensuring that only runoff meeting established standards is released. Intervention at the policy and urban planning levels is more efficient than attempting to rectify issues through design guidelines alone. Therefore, government agencies and relevant stakeholders must prioritize proactive policy and planning decisions that will help cities respond more effectively and resiliently to the challenges posed by climate change.

From the perspective of urban planning and design, this study argues that localized green infrastructure strategies should be coordinated with large-scale landscape planning to enable a more forward-looking and resilient approach to urban water management. Such integration requires cross-scalar and multi-sectoral collaboration to address the complex dynamics of urban hydrological systems.

First, a data-driven planning approach is essential to enhance understanding and responsiveness to natural hydrological processes. By utilizing multi-source data such as remote sensing, digital elevation models (DEMs), and hydrological simulations, planners can better analyze river dynamics and sediment transport patterns. This scientific basis supports more informed decisions regarding urban spatial layout and future development. Second, water-related spatial planning should be conducted at the watershed scale. Analytical results concerning flood-prone areas and sediment flow pathways can inform the spatial configuration of urban blue-green infrastructure networks. These insights can guide the rational allocation of ecological buffers, flood detention zones, and wetland systems. Such measures serve as a key reference for urban master planning and regional landscape structure design, ensuring that urban development—especially in riparian areas—respects natural hydrological processes and reduces the likelihood of conflict with natural hazards. In this way, the comprehensive management of water environments can move beyond site-specific engineering solutions toward an ecologically integrated planning framework, fostering a more harmonious coexistence between urban growth and natural systems.

6 Conclusion

This study explores the application of ABC Waters design features in riparian spaces. Different types of swales are initially selected and positioned based on runoff routes. Concurrently, the embankment design is optimized by integrating the Slope Terrain Index (STI) with runoff paths to reduce river sedimentation caused by soil erosion. Following this, an analysis of the sources of river pollutants in the surrounding area informs the selection of appropriate vegetation for the swale system, aiming to improve water quality and enhance site biodiversity. Finally, interactive landscape elements are incorporated into the design, leveraging the site's topography to strengthen connections between the area and the community. Unlike conventional studies that treat river flood management and sediment control as separate issues, this research adopts a holistic approach using ABC Waters design guidelines to develop a comprehensive solution. This approach provides a reference case for integrating floodwater and sediment management in riparian areas. It improves urban waterfront spaces' resilience, environmental quality, and community engagement. This study will raise awareness and encourage support for flood management and sediment control efforts, creating more livable and resilient urban environments.

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